

WEEK 12 — FORENSIC INVESTIGATION

PART A: Disk Triage & Artefact Hunting

Cyberster Blue Team Internship — Week 12 — Days 81–84
Subject: Jo (M57.biz) | Image: jo-2009-11-24.E01 | Timezone: UTC +0:00

Case	M57.biz — Jo Insider Threat Investigation
Image	jo-2009-11-24.E01
MD5	274ada9b6b120f9b7a2192c27e115244
OS	Microsoft Windows XP Service Pack 3 (x86)
Computer	M57-JO
Owner	JO — M57.biz
Timezone	UTC +0:00 (all timestamps UTC)
Tools	Autopsy 4.23.0 — full ingest modules
Acquired	2011-01-13 11:54:36 UTC (Linux — ewfacquire v.20100226)
USB Images	jo-favorites-usb-2009-12-11.E01 jo-work-usb-2009-12-11.E01
RAM Image	jo-2009-11-16.mddramimage
Focus	System overview, user accounts, USB events, downloads, LNK artefacts, shell bags

1. System Overview & Image Verification

The forensic image jo-2009-11-24.E01 was loaded into Autopsy with the timezone set to UTC +0:00. Ingest modules — including Recent Activity, Hash Lookup, Keyword Search, and File Type Identification — were executed before analysis commenced. The image is 15.38 GB in size and was acquired from a Linux system using ewfacquire.

Field	Value
Image Name	jo-2009-11-24.E01
Type	E01 (EnCase Expert Witness Format)
Size	15,382,241,280 bytes (~15.38 GB)
MD5	274ada9b6b120f9b7a2192c27e115244
Sector Size	512 bytes
Acquired	2011-01-13 11:54:36 UTC
Acq. Software	ewfacquire v.20100226 (Linux)
Computer Name	M57-JO
OS	Microsoft Windows XP SP3 (x86)
Registered Owner	JO
Organisation	M57.biz
Product ID	76487-027-5250835-22726
Windows Path	C:\WINDOWS
Temp Directory	%SystemRoot%\TEMP

2. User Accounts

SID (abbreviated)	Login Name	Scope	Realm / Notes
S-1-5-21-...-500	administrator	Local	Built-in administrator
S-1-5-21-...-1003	jo	Local	Primary user — all suspicious artefacts within this profile
S-1-5-18	SYSTEM	Local	NT AUTHORITY — system account
S-1-5-19	LOCAL SERVICE	Local	NT AUTHORITY
S-1-5-20	NETWORK SERVICE	Local	NT AUTHORITY

■ NOTE

The primary investigative account is **jo** (SID ending -1003). All suspicious artefacts — LNK files, shell bags, downloads, desktop content, and USB activity — are located within **C:\Documents and Settings\Jol**.

3. Key Installed Programs

Program	Version	Install Date	Investigative Significance
Python 2.6.4	2.6.4150	2009-11-23	Required runtime for patentauto.py — installed same day as s
Mozilla Firefox 3.5.5	3.5.5	2009-11-20	Primary browser — used by patentauto.py via MozRepl on por
Java (TM) 6 Update 16	6.0.160	2009-11-20	Required to execute DIIT steganography tool (diit-1.5.jar)
OpenOffice.org 3.1	3.1.9420	2009-11-20	Office suite — spawned cmd.exe processes in memory (see P
AVG 9.0	9.0	2009-11-20	Antivirus — high run count (104) during investigation
QuickTime 7.65	7.65.17.80	2009-11-24	Installed on acquisition date — media player

■ CAUTION

Python 2.6.4 was installed on **2009-11-23** — the same day patentauto.py was created. Firefox 3.5.5 was already present and was leveraged by the script via the MozRepl extension on localhost port 4242. Java 6 was required for DIIT.

4. LNK Artefact Analysis — Recent Folder

The following LNK shortcut files were recovered from C:\Documents and Settings\Jo\Recent\ and confirm user interaction with each target file or folder.

LNK File	Target Path	Last Accessed (UTC)	Significance
web.lnk	Desktop\web\	2009-11-23 21:55:46	Folder accessed — contains patentauto.py and URL
Pics.lnk	Desktop\Pics\	2009-11-23 18:24:51	Folder accessed — contains DIIT and test images
HighQuality.lnk	Desktop\Pics\HighQuality\	2009-11-24 22:14:30	High-res image folder accessed
patentauto.lnk	Desktop\web\patentauto.py	2009-11-23 21:55:45	Patent script opened / executed
diit-1.5.lnk	Desktop\Pics\diit-1.5.jar	2009-11-23 18:24:50	DIIT steganography tool accessed
testHide.lnk	Desktop\Pics\testHide.bmp	2009-11-23 18:25:53	Steganography test image accessed (93 s after DIIT)
Downloads.lnk	My Documents\Downloads\	2009-11-24 22:01:54	Downloads folder accessed
STU.lnk	My Documents\Downloads\STU.doc	2009-11-24 22:01:53	Classified DoC document accessed
DSC00065.lnk	Desktop\Pics\HighQuality\DSC00065.JPG	2009-11-24 22:14:30	High-res photograph accessed
DSC00070.lnk	Desktop\Pics\HighQuality\DSC00070.JPG	2009-11-24 22:14:45	High-res photograph accessed

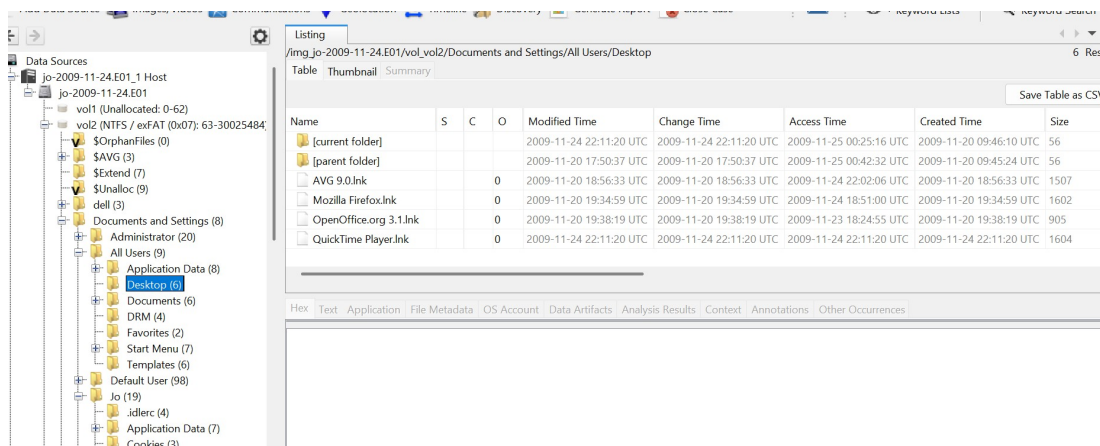


Figure 1 — Desktop contents showing Pics and web folders

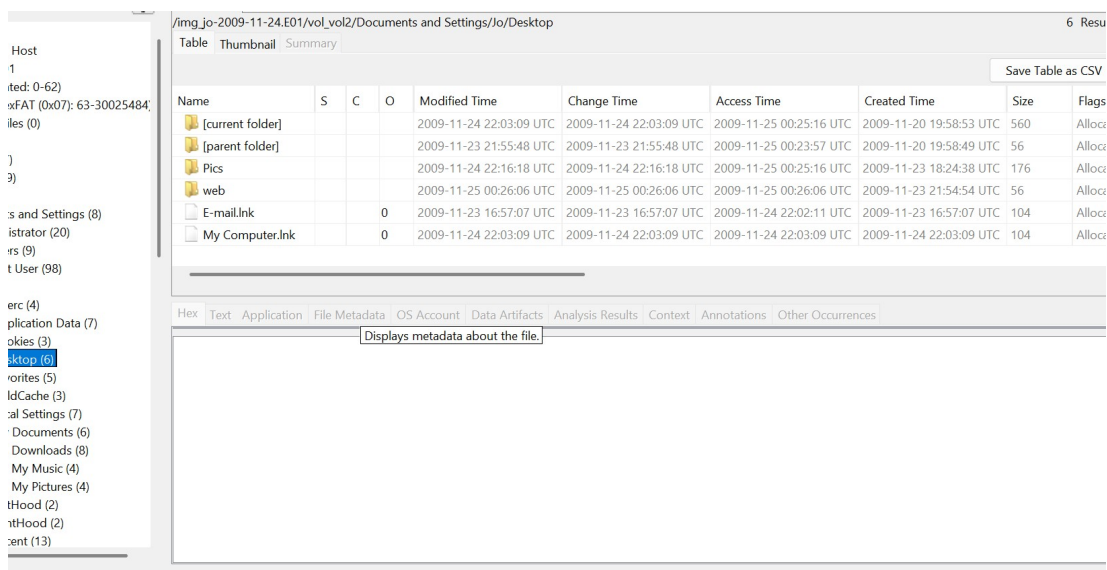


Figure 2 — Desktop file listing in Autopsy

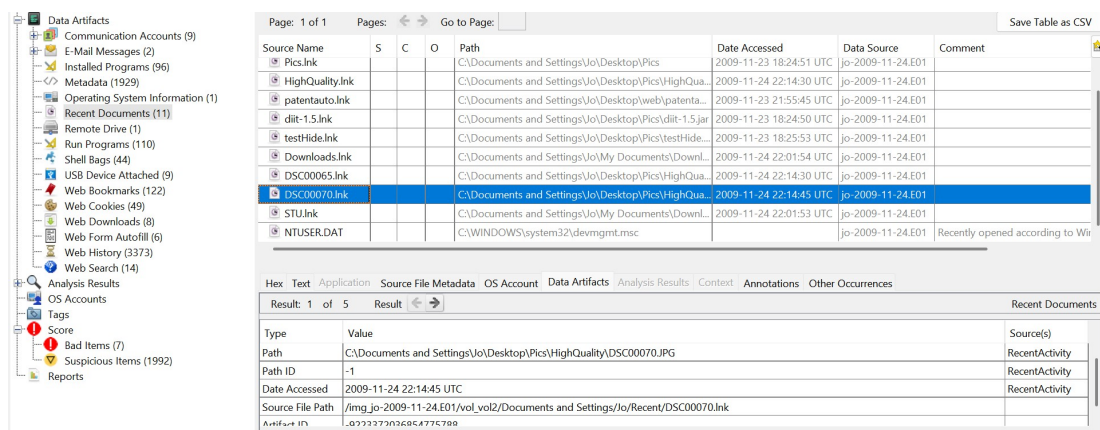


Figure 3 — Recent folder LNK artefacts in Autopsy

5. Downloads Analysis

File	Source	Downloaded (UTC)	MD5	Significance
python-2.6.4.msi	python.org	2009-11-23 18:20:12	2e2b60ae...	Python runtime — enables patentauto.py
STU.doc	easc.noaa.gov	2009-11-24 21:18:18	929af3c7...	DoC SECRET briefing — keyword hit: 'SE
QuickTimeInstaller.exe	apple.com CDN	2009-11-24 22:08:36	5c9ecf92...	Media player installer — downloaded day

■ CRITICAL FINDING

STU.doc was downloaded from a US Department of Commerce / NOAA government server (easc.noaa.gov). The file contains a complete STU-III classified telecommunications security briefing authored by the DoC Office of Security. Autopsy's keyword search flagged it as 'Likely Notable' with the keyword **SECRET**. There is no evident legitimate business reason for a M57.biz employee to possess this document.

Name	S	C	O	Modified Time	Change Time	Access Time	Created Time
[parent folder]				2009-11-23 18:20:14 UTC	2009-11-23 18:20:14 UTC	2009-11-25 00:25:16 UTC	2009-11-20 19:58:51 U
python-2.6.4.msi			1	2009-11-23 18:23:03 UTC	2009-11-23 18:23:03 UTC	2009-11-24 01:29:33 UTC	2009-11-23 18:20:14 U
python-2.6.4.msi.Zone.Identifier			1	2009-11-23 18:23:03 UTC	2009-11-23 18:23:03 UTC	2009-11-24 01:29:33 UTC	2009-11-23 18:20:14 U
QuickTimeInstaller.exe			0	2009-11-24 22:08:50 UTC	2009-11-24 22:09:05 UTC	2009-11-24 22:09:05 UTC	2009-11-24 22:08:37 U
QuickTimeInstaller.exe.Zone.Identifier			1	2009-11-24 22:08:50 UTC	2009-11-24 22:09:05 UTC	2009-11-24 22:09:05 UTC	2009-11-24 22:08:37 U
STU.doc			1	2009-11-24 22:01:53 UTC	2009-11-24 22:01:53 UTC	2009-11-24 22:01:53 UTC	2009-11-24 22:01:51 U
STU.doc.Zone.Identifier			1	2009-11-24 22:01:53 UTC	2009-11-24 22:01:53 UTC	2009-11-24 22:01:53 UTC	2009-11-24 22:01:51 U

Figure 4 — Downloads folder: STU.doc, Python installer, QuickTime

6. USB Device Evidence

Device	Make / Model	Device ID	Connected (UTC)	Significance
Flash Drive	Alcor Micro Corp.	51491E64	2009-11-23 21:54:44	Connected 51 s before patentauto.py acces
Unknown USB	Vid_0000&Pid_7777	2B9ECCFF	2009-11-24 22:01:49	Connected during STU.doc / image access
Logitech Mouse	Logitech Optical	5&2f6f8af0...	2009-11-24 01:28:37	Standard peripheral
Dell Keyboard	Dell L100	5&2f6f8af0...	2009-11-20 17:55:41	Standard peripheral

■ CRITICAL FINDING

The **Alcor Micro Flash Drive** was connected at 21:54:44 on 2009-11-23 — 51 seconds before patentauto.lnk was created at 21:55:45, and 32 seconds before the patentauto.py access event. The **unknown device** (Vid_0000&Pid_7777) was connected at 22:01:49 on 2009-11-24 — during the session when STU.doc and high-quality images were accessed. Both events are temporally consistent with data staging or exfiltration activity.

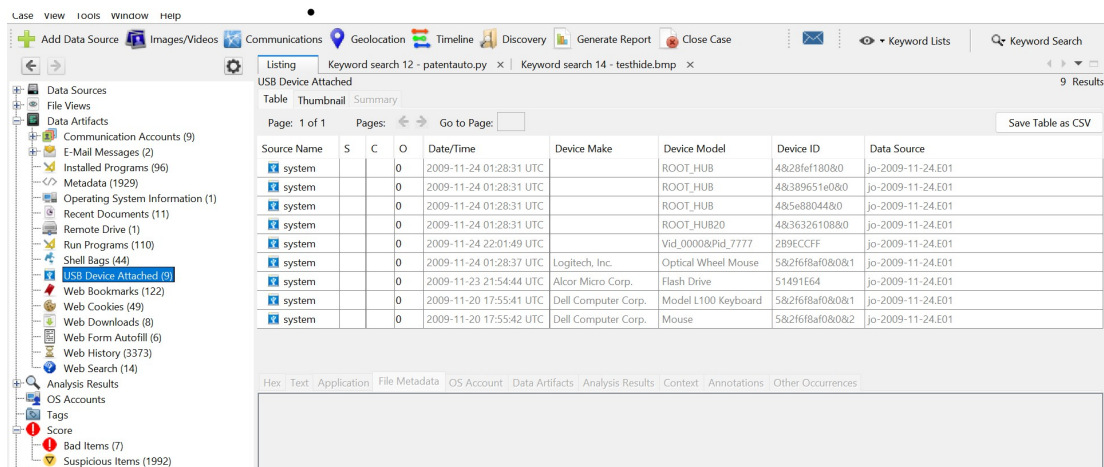


Figure 5 — USB device attachment events in Autopsy

7. Shell Bag Analysis

Shell bags record folder navigation history independently of LNK shortcuts. Entries in NTUSER.DAT at Software\Microsoft\Windows\ShellNoRoam\BagMRU confirm Jo browsed into the following directories:

Item	Shell Bag Key	Significance
patentauto.py	ShellNoRoam\Bags\10\Shell	Patent automation script — browsed to and accessed
patentterms.txt	ShellNoRoam\Bags\10\Shell	Search terms — input file for patentauto.py
urls.txt	ShellNoRoam\Bags\10\Shell	URL list — scraping targets for patent data
urlscopyright.txt	ShellNoRoam\Bags\10\Shell	Copyright-related patent URLs
urlspatents.txt	ShellNoRoam\Bags\10\Shell	Direct patent record URLs
urlspersona.txt	ShellNoRoam\Bags\10\Shell	Identity/persona patent research URLs
urlstime_machine.txt	ShellNoRoam\Bags\10\Shell	Internet Archive / historical patent sources
Pics folder	Shell\Bags\1\Desktop	Accessed 2009-11-24 17:16:20 — steganography folder
web folder	Shell\Bags\1\Desktop	Accessed 2009-11-24 19:26:08 — patent script folder

The web folder contained a structured collection of patent intelligence files — a Python script plus six categorised URL/term input files. This organisation reflects deliberate, systematic data collection rather than casual browsing.

8. Email Evidence & Work Documents

Outlook Express (MSIMN.EXE) was installed and shows a prefetch run count of 4, last executed 2009-11-24 22:02:13 UTC. A spam email was identified in Jo's mailbox. Work papers and document artefacts were also recovered from the profile.



Listing

Keyword search 3 - zip x

Keyword search

296 Results

Table

Thumbnail

Summary

Save Table as CSV

Name	Keyword Preview	Location	Modified Time	Check
square-yellow.zip	square=yellow.zip	/img_jo-2009-11-24.E01/vol_vol2/Program Files/Ope...	2009-01-27 21:30:54 UTC	200
avgwdsvc_idp_SYSTEM.log	update\prepare\temp\internalist.zip.prepare, timeo...	/img_jo-2009-11-24.E01/vol_vol2/Documents and S...	2009-11-25 00:46:15 UTC	200
square-yellow.zip-slack	square=yellow.zip=slac	/img_jo-2009-11-24.E01/vol_vol2/Program Files/Ope...	2009-01-27 21:30:54 UTC	200
internalList.zip	=internalist.zip	/img_jo-2009-11-24.E01/vol_vol2/Documents and S...	2009-11-20 18:55:43 UTC	200
msobshel.htm	ed = true; =g.zipcode.disabled= = true; g.	/img_jo-2009-11-24.E01/vol_vol2/WINDOWS/syste...	2008-04-14 12:00:00 UTC	200
internalList.zip-slack	=internalist.zip=slac	/img_jo-2009-11-24.E01/vol_vol2/Documents and S...	2009-11-20 18:55:43 UTC	200
_REGISTRY_MACHINE_SOFTWARE	-08002b30bfefjnk f=zip=content typeapplic	/img_jo-2009-11-24.E01/vol_vol2/System Volume In...	2009-11-20 18:47:34 UTC	200
NTOSBOOT-800DFAAD.pf	ata\avg9\ids\config\userlist.zip.bak\device\harddisk...	/img_jo-2009-11-24.E01/vol_vol2/WINDOWS/Prefet...	2009-11-24 01:30:14 UTC	200
REGISTRY USER NTUSER S-1-5-21-606747145-15	wmx wml wrl wwwzineennuwithnroonrks wa	/img_jo-2009-11-24.E01/vol_vol2/System Volume In	2009-11-20 18:47:33 UTC	200

Hex

Text

Application

File Metadata

OS Account

Data Artifacts

Analysis Results

Context

Annotations

Other Occurrences

Basic Properties

Login: administrator

Full Name:

Address: S-1-5-21-606747145-1547161642-1644491937-500

Type:

Creation Date:

Object ID: 7

jo-2009-11-24.E01_1 Host Details

Figure 9 — internalList.zip found in AVG download manager temp folder